Phone Number Mnemonics

Topic.

Trie, dynamic programming, shortest decomposition, answer reconstruction.

Statement.

Given a phone number and a dictionary of words, find the shortest sequence of words that corresponds to the phone number.

Each digit can be represented by several letters according to the phone keypad mapping:

$$\begin{array}{c|c} 1 & ij \\ 2 & abc \\ 3 & def \\ 4 & gh \\ 5 & kl \\ 6 & mn \\ 7 & prs \\ 8 & tuv \\ 9 & wxy \\ 0 & oqz \end{array}$$

Each word corresponds to a unique sequence of digits.

The task is to represent the given phone number as a sequence of dictionary words with the minimum possible number of words.

Input.

The input consists of several tests.

The first line of each test contains the phone number. The number has length at most 100.

The second line contains the number of dictionary words.

Each of the following lines contains one word consisting of lowercase English letters. Each word has length at most 50.

The input ends with a line containing

$$-1.$$

Output.

For each test, print the shortest sequence of words corresponding to the phone number.

Words must be separated by single spaces.

If there is no such sequence, print

`No solution.`

Implementation.

The implementation is provided separately in the file

`solution.cpp`.